

Element F: Consideration of Design Viability

The stakeholder's problem was including special-need students and helping them participate in large-size physical education classes. The main problem was that these students had limited motor skills and most of the time could not use a large part of their body. As a result, they were confined to walkers and wheelchairs which made it difficult to participate in typical physical activities. Frisbees help create an entertaining environment and can be easily used by any student. The proposed solution of a frisbee launcher that was developed aimed at fulfilling all of the stakeholder's needs while still being entertaining for the students. The frisbee launcher was designed to launch a frisbee with the help of motors rather than a student's arm. This way, a game of frisbee can still be played without having to move body parts around, which can be hard or impossible for these students. The Frisbee launcher contains a subsystem in which two running motors are connected by a belt to create a pulley system. The two motors have a flywheel attached to them which causes the flywheels to spin at a high rate of speed. As a result, the frisbee that passes by gets a thrust of power and will shoot out similar to a human-arm throwing one.

The use of viable materials such as wood, pulleys, motors and flywheels can facilitate the development of a realistic prototype for the user. These materials are easily attainable and can be implemented into our design to allow the frisbee to fly at the desired distance. Similar prototypes have been built using wood as the base and a motor system to make the flywheel spin. However, in these prototypes, the motor was directly underneath the flywheel, while the proposed design to the user requires a pulley system, which uses a belt to connect the motor to each axle of the two flywheels. These materials were also easily attainable and allowed for them to be efficiently used into the development of the prototype.

Citations:

Disc Golf Throwing Robot. June 2023. *Cal Poly*, digitalcommons.calpoly.edu/cgi/viewcontent.cgi?article=1835&context=mesp. Accessed 2 June 2024.

The initial design requirements sought for a simple, easy-to-use, and safe prototype that can engage students with limited abilities in physical activities. However, while having only one flywheel would be sufficient to meet the user's needs, the proposed prototype consists of two flywheels which can shoot the frisbee at a larger distance. In addition, instead of having just one set speed to shoot the frisbee at, speed of the motor can be adjusted in order to improve the feasibility of the prototype. Using Arduino, we can connect the motor to the pins and manipulate the speed by decreasing the torque of the motor. The solution will also be able to adjust by changing the angle at which the frisbee can be launched. By changing the

In order to ensure the extra aspects of the prototype were compelling, a professor at George Mason University reviewed our design. While the professor had supported our ideas for a double flywheel system, he also had concerns about the added weight of the whole mechanism. In order to add the second flywheel, there would be many other parts such as the extra material to mount the motor and the pulleys. However, the added flywheel would allow the frisbee to shoot farther and engage the students even more. Furthermore, the extra feature of changing the motor speed can also manipulate the distance the frisbee will fly. This allows for the prototype to be more flexible for the user and can be used to cater the needs for many different students.